Image Plagiarism Detection with Equivariant Encoders

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Abstract

Detecting plagiarized images requires robustness to rotations, reflections, and shifts. We ask whether geometric inductive bias in the encoder outperforms data augmentation for this robustness. We train a Siamese network with equivariant ViT encoders under three symmetry groups—D_4, $\mathbb{Z}^2$, and SE(2)—with an L^2 contrastive regularizer, giving the first controlled three-group comparison in one framework. Evaluation on a balanced DomainNet protocol uses FPR and Recall. Applications include plagiarism screening in journalism, academia, and design.

1 Introduction

Image plagiarism detection asks whether two images x and x' depict the same underlying content despite geometric or photometric transformations applied to one of them. The central research question of this paper is: *does building geometric symmetry directly into the encoder architecture improve plagiarism detection over relying on data augmentation alone?* We study this question systematically by comparing five encoder configurations across three symmetry groups—D_4, $\mathbb{Z}^2$, and $SE(2)$—within a single siamese framework, evaluating each with and without an L^2 contrastive regularizer on a balanced cross-domain benchmark.

The object of study is pairs of natural images related by transformations drawn from D_4 (rotations by multiples of 90° and reflections), $\mathbb{Z}^2$ (integer pixel shifts), or $SE(2)$ (arbitrary rotations and translations), compounded with photometric perturbations such as brightness jitter, contrast scaling, and JPEG compression. Standard Vision Transformers (ViTs) [1] lack geometric invariance by construction: their patch-embedding tokenization is not equivariant to rotations or reflections, and their self-attention mechanism treats spatial position through learned positional encodings that generalize poorly outside the training distribution. Practitioners therefore rely on geometric data augmentation to approximate robustness at inference time; however, augmentation only stochastically covers the orbit of a transformation group, inflates training cost, and provides no invariance guarantee on unseen perturbations. Moreover, scaling model size does not resolve the underlying limitation—recent evidence shows that spatial and geometric reasoning fails to improve with scale alone [2]. Consequently, there is a principled motivation to embed the desired symmetry as an inductive bias in the architecture itself, following the program of geometric deep learning [3, 4].

Formally, an encoder $\varphi_\theta : \mathcal{X} \to \mathcal{Z}$ is *G-equivariant* if $\varphi_\theta(g \cdot x) = \rho(g)\, \varphi_\theta(x)$ for all $g \in G$ and all images x, where ρ is a representation of G on the embedding space $\mathcal{Z}$; it is *G-invariant* when ρ is the trivial representation, i.e. $\varphi_\theta(g \cdot x) = \varphi_\theta(x)$. In plagiarism detection the siamese head compares embeddings $z = \varphi_\theta(x)$ and $z' = \varphi_\theta(x')$, so invariance of φ_θ directly implies that geometric variants of the same image collapse to the same embedding without any augmentation. Full theoretical treatment of these properties is deferred to Section 4.

Recent work has addressed geometric robustness in vision transformers from several directions. Nordström et al. [5] introduce the Octic ViT, which enforces exact D_4-equivariance through group-structured weight sharing and reduces FLOPs by 40% relative to a standard ViT of equal expressivity.

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Table 1: Comparison of approaches to geometric robustness in image matching.

Method	Group	Exactness	Strengths	Weaknesses
Dorin et al. [9]	None	Approximate	Flexible; strong features	No invariance guarantee
Octic ViT [5]	D_4	Exact	40% FLOPs reduction	Discrete; 90° only
Shift-Eq. ViT [6]	$\mathbb{Z}^2$	Exact	Exact shift invariance	No rotation equivariance
Harmformer [7]	$SE(2)$	Exact	Arbitrary angles + shifts	Higher compute cost

Rojas-Gomez et al. [6] design a Shift-Equivariant ViT that achieves exact $\mathbb{Z}^2$-invariance via anti-aliased downsampling and circular padding, but without rotational equivariance. Karella et al. [7] propose Harmformer, which lifts transformers to the continuous group $SE(2)$ using harmonic feature maps, providing equivariance to arbitrary in-plane rotations and translations at the cost of higher compute. On the side of learned features and plagiarism-specific architectures, Oquab et al. [8] demonstrate that DINOv2 pre-training on large unlabeled image sets produces highly transferable representations, and Dorin et al. [9] establish a siamese ViT baseline for pairwise plagiarism detection with false positive rate (FPR) as the primary evaluation metric [10].

We propose a siamese plagiarism detector with G-equivariant ViT encoders, providing exact invariance under D_4, $\mathbb{Z}^2$, or $SE(2)$ without geometric augmentation. The novelty relative to Dorin et al. [9] is twofold: (i) an L^2 contrastive regularizer that directly penalizes the Euclidean distance between embeddings of genuine pairs while separating impostor pairs, and (ii) a controlled augmentation ablation that compares geometric augmentation against equivariant inductive bias across three symmetry groups in one unified framework. Concretely, we execute five encoder configurations—standard ViT with augmentation, DINOv2-pretrained ViT, Octic ViT (D_4), Shift-Equivariant ViT ($\mathbb{Z}^2$), and Harmformer ($SE(2)$)—all initialized with DINOv2 weights on ImageNet-1K where architecturally compatible, and each trained in two conditions: with and without the L^2 contrastive regularizer.

The experiment comprises 10 runs (5 encoders $\times$ {with, without} L^2 regularizer) evaluated on a class-balanced subset of DomainNet spanning six visual domains. Primary metrics are FPR and Recall at a fixed decision threshold; secondary metric is F1. This design isolates the contribution of each inductive bias and directly tests whether exact geometric invariance reduces false positives relative to an augmentation-only baseline. Table 1 summarises the methods compared.

The remainder of this paper is organized as follows. Section 2 reviews related work on equivariant networks, siamese architectures, and plagiarism detection. Section 3 formalizes the detection task and evaluation protocol. Section 4 develops the theoretical background on group equivariance. Section 5 describes the siamese framework, encoder configurations, and the L^2 contrastive regularizer. Section 6 presents experimental results and ablations. Section 7 concludes with limitations and future directions.

2 Related Work

Siamese detectors for image matching and plagiarism. Bromley et al. [10] introduced siamese networks for signature verification, demonstrating that a shared-weight twin architecture with a learned distance metric generalises naturally to any pairwise similarity task. Classical near-duplicate detection relied on compact local descriptors combined with geometric verification [11], an approach that breaks under large photometric or domain shifts. Dorin et al. [9] applied a siamese ViT with contrastive loss to image plagiarism detection on DomainNet, reporting false positive rate (FPR) as the primary metric and using a cross-batch pairing strategy; our work extends their protocol with equivariant encoders and an L^2 contrastive regularizer that directly controls the geometry of the embedding space.

Equivariant and invariant architectures. Cohen and Welling [4] introduced G-CNNs that encode discrete symmetries into convolutional filters via group-structured weight sharing; Weiler and Cesa [12] generalised this approach to steerable E(2)-equivariant CNNs, providing a unified treatment of continuous rotation and reflection groups. Bronstein et al. [3] provide a unifying geometric deep learning framework in which equivariance is cast as an inductive bias that improves sample efficiency and generalisation across domains. These architectures have been evaluated primarily on

classification and segmentation tasks; we are the first to apply equivariant encoders to pairwise image similarity in a siamese plagiarism-detection framework.

Equivariant Vision Transformers. Nordström et al. [5] replaced standard ViT layers with D_4-equivariant variants, achieving a 40% FLOPs reduction at accuracy comparable to the baseline ViT. Rojas-Gomez et al. [6] achieve exact shift-equivariance via adaptive polyphase sampling—introducing A-token, A-WSA, A-PMerge, and A-RPE modules—without adding learnable parameters. Karella et al. [7] combine circular harmonic filters with transformer blocks to obtain continuous SE(2)-equivariance, extending roto-translation symmetry to arbitrary angles. No prior work evaluates any of these architectures on pairwise image similarity or plagiarism detection, leaving open the question of whether geometric inductive bias reduces false positives.

Self-supervised ViT pre-training and contrastive learning. DINOv2 [8] produces strong, transferable visual features via self-supervised learning on large unlabeled image collections; we use it as a common pre-training recipe across all encoders to ensure that performance differences reflect architectural choices rather than pre-training data. Chopra et al. [13] proposed contrastive loss for discriminative metric learning, showing that minimising intra-class distance while maximising inter-class distance yields strong verification performance. Hadsell et al. [14] introduced the L^2 contrastive formulation we adopt, motivated by its isometry invariance: the loss depends only on Euclidean distances in embedding space, making it robust to rigid transformations of the latent representation.

3 Problem Statement

1. General problem. Given a pair of images $(x_1, x_2) \in \mathcal{X} \times \mathcal{X}$, the task is binary classification: predict $y \in \{0, 1\}$ where $y = 1$ indicates that x_2 is a plagiarised derivative of x_1. Plagiarism is operationalised as the composition of a geometric transformation $g \in G$ with a photometric perturbation $\pi \in G_{\text{photo}}$ (colour jitter, Gaussian blur, additive noise), yielding $x_2 = (g \circ \pi)(x_1)$. The detector must correctly identify such pairs regardless of which transformation was applied.

2. Sample set. The labelled dataset is

$$\mathcal{D} = \left\{(x_1^{(i)},\, x_2^{(i)},\, y^{(i)})\right\}_{i=1}^{N},$$

where $y^{(i)} \in \{0, 1\}$ and every image $x^{(i)} \in \mathcal{X} = [0, 1]^{3\times 224\times 224}$. Each sample is a triple: a query image, a candidate image, and a binary plagiarism label. The dataset is balanced across positive and negative pairs and stratified by visual domain.

3. Statistical hypothesis. Positive pairs are drawn from the joint distribution

$$p_+(x_1, x_2) = p(x_1)\, p_T(x_2 \mid x_1),$$

where $p_T(x_2 \mid x_1) = \int_{G\times G_{\text{photo}}} \delta(x_2 - (g \circ \pi)(x_1))\, d\mu(g, \pi)$ is the mixture of transformation kernels over $G \cup G_{\text{photo}}$. Negative pairs are obtained by sampling x_1 and x_2 independently from the marginal $p(x)$, so $p_-(x_1, x_2) = p(x_1)\, p(x_2)$. The detector must distinguish the two regimes on held-out data.

4. Measurement conditions. All experiments use the DomainNet dataset [15], which spans six visually distinct domains: real photograph, painting, clipart, quickdraw, infograph, and sketch. Images are resized to 224×224 pixels in 8-bit RGB and normalised per-channel using ImageNet mean and standard deviation. This multi-domain structure enables evaluation of generalisation across severe appearance shifts.

5. Sample restrictions. Positive pairs consist of a source image x and its transformed copy $(g \circ \pi)(x)$, where $g \sim \text{Uniform}(G)$ and $\pi \sim p_{\text{photo}}$. Negative pairs are constructed from two independently sampled images drawn from the *same* DomainNet domain, making them hard negatives with shared visual statistics. The evaluation set contains 1200 pairs in total: 100 positives and 100 negatives for each of the six domains, ensuring balanced coverage.

6. Model. The full model is $f_\theta : \mathcal{X} \times \mathcal{X} \to [0,1]$, decomposed as

$$f_\theta(x_1, x_2) = \sigma\big(h_\psi(\varphi_\theta(x_1),\ \varphi_\theta(x_2))\big), \tag{1}$$

where $\varphi_\theta : \mathcal{X} \to \mathbb{R}^d$ is the encoder with shared weights, h_ψ is the fusion module, and σ denotes the sigmoid function. The siamese design ensures that both images are mapped to the same embedding space before comparison.

7. Class restrictions. The encoder φ_θ must be G-equivariant with a G-invariant pooling step so that the resulting embedding satisfies $\varphi_\theta(g \cdot x) = \varphi_\theta(x)$ for all $g \in G$. This invariance guarantees that geometric variants of the same image collapse to identical embeddings without data augmentation. Additionally, the fusion module h_ψ must be symmetric: $h_\psi(z_1, z_2) = h_\psi(z_2, z_1)$, ensuring that the plagiarism score does not depend on the ordering of the two inputs.

8. Loss / criterion. Training minimises the composite objective

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}} + \lambda\, \mathcal{L}_{\mathrm{contr}}^{L^2}, \tag{2}$$

where $\mathcal{L}_{\mathrm{BCE}}$ is weighted binary cross-entropy with $w_+ = 0.3$ and $w_- = 0.7$, and the contrastive regulariser is

$$\mathcal{L}_{\mathrm{contr}}^{L^2}(z_1, z_2, y) = y\, \|z_1 - z_2\|_2^2 + (1-y)\ \max\big(0,\ m - \|z_1 - z_2\|_2\big)^2. \tag{3}$$

Setting $\lambda = 0$ recovers the variant trained without contrastive regularisation, which serves as a baseline to isolate the contribution of the geometric loss term.

9. Cross-validation procedure. Generalisation is estimated via stratified five-fold cross-validation over the DomainNet evaluation pool, with stratification by domain to ensure that every fold contains examples from all six domains. Hyper-parameters are selected on the validation fold, and test metrics are averaged across folds. This procedure is applied identically to all encoder configurations to ensure comparability.

10. Solution restrictions. All experiments are conducted on a single GPU with at most 48 GB of VRAM, and fine-tuning is limited to at most 30 epochs. The encoder backbone φ_θ is frozen throughout training; only the fusion head h_ψ is updated. These constraints reflect a realistic resource budget and ensure that differences in performance reflect architectural choices rather than training cost.

11. External quality criteria. The primary metrics are **FPR** (false positive rate at the threshold that maximises F1 on the validation fold) and **Recall** reported per transformation class: colour jitter (CJ), Gaussian noise (GN), Gaussian blur (GS), rotation by 90° (R90), 180° (R180), 270° (R270), watermark (WM), and combinations thereof. Secondary metrics are F1 score and t-SNE separability of embeddings z on DomainNet. Stratified recall directly measures whether a given encoder is robust to specific transformation types.

12. Argmin. The optimal encoder and fusion parameters are given by

$$\hat{\theta} = \arg\min_{\theta}\ \mathbb{E}_{(x_1,\, x_2,\, y)\, \sim\, \mathcal{D}}\, \mathcal{L}(\theta). \tag{4}$$

In practice, this expectation is approximated by the empirical mean over mini-batches, and $\hat{\theta}$ is taken as the checkpoint with the lowest validation loss across the five cross-validation folds.

4 Preliminaries: Equivariance

4.1 Group Actions and Representations

Definition 1 (Group action)**.** *A* left action *of a group G on a set $\mathcal{X}$ is a map $G \times \mathcal{X} \to \mathcal{X}$, $(g, x) \mapsto g \cdot x$, satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$, $x \in \mathcal{X}$.*

Definition 2 (Group representation)**.** *A* (linear) representation *of G on a vector space V is a group homomorphism $\rho\colon G \to \mathrm{GL}(V)$.*

Representations describe concretely how G acts on feature spaces: given a representation ρ, each group element g is assigned an invertible linear transformation $\rho(g)$ of V, and the group multiplication is preserved by $\rho(gh) = \rho(g)\rho(h)$.

4.2 Equivariance and Invariance

Definition 3 (G-Equivariance)**.** *A map $\varphi\colon \mathcal{X} \to \mathcal{Y}$ is G-*equivariant *with respect to representations $\rho_{\mathcal{X}}$ on $\mathcal{X}$ and $\rho_{\mathcal{Y}}$ on $\mathcal{Y}$ if*

$$\varphi\big(\rho_{\mathcal{X}}(g) \cdot x\big) = \rho_{\mathcal{Y}}(g) \cdot \varphi(x) \quad \forall\, g \in G,\ x \in \mathcal{X}.$$

Definition 4 (G-Invariance)**.** *φ is G-*invariant *if $\rho_{\mathcal{Y}}$ is the trivial representation: $\varphi(\rho_{\mathcal{X}}(g) \cdot x) = \varphi(x)$ for all $g \in G$.*

Invariance is therefore a special case of equivariance in which the output space carries only the trivial action of G. For plagiarism detection, a G-invariant encoder φ_θ guarantees $\varphi_\theta(g \cdot x) = \varphi_\theta(x)$ for every $g \in G$, so embeddings are identical regardless of which group element was applied to the source image. This eliminates the need for group-specific augmentation at training time and provides an exact invariance guarantee at inference time.

4.3 Irreducible Representations and Schur's Lemma

Lemma 1 (Schur)**.** *Let $\rho_1\colon G \to \mathrm{GL}(V_1)$ and $\rho_2\colon G \to \mathrm{GL}(V_2)$ be irreducible representations of a group G. Any G-equivariant linear map $\phi\colon V_1 \to V_2$ is either the zero map or an isomorphism.*

As a consequence of Schur's lemma, G-equivariant linear maps between fully decomposed representations are block-diagonal with respect to the irreducible decomposition: each block acts within one irreducible component and cannot mix distinct irreducible types. This constraint significantly reduces the parameter space of equivariant layers compared to unconstrained linear maps. It directly motivates the architecture of Octic ViT [5], where each transformer layer operates independently on each irreducible component of D_4, enforcing the block-diagonal structure by construction.

4.4 Symmetry Groups in This Work

The dihedral group D_4. $D_4 = \langle r, s \mid r^4 = s^2 = e,\ srs = r^{-1}\rangle$ is the symmetry group of the square, of order 8. It acts on images by rotating and reflecting the spatial axes by multiples of 90°. D_4 has five irreducible representations over $\mathbb{R}$: four one-dimensional (A_1, A_2, B_1, B_2) and one two-dimensional (E); any D_4-representation decomposes as $V \cong \bigoplus_i n_i V_i$ over these five types.

The discrete translation group $\mathbb{Z}^2$. The group $(\mathbb{Z}^2, +)$ acts on an $H \times W$ image by cyclic pixel shifts: $(t_{a,b} \cdot x)[i, j, c] = x[(i + a) \bmod H,\ (j + b) \bmod W,\ c]$. It is relevant when plagiarists crop and re-centre images. Unlike D_4, $\mathbb{Z}^2$ is infinite (or large-but-finite on discrete images), making exact equivariance more challenging to enforce in the ViT patch-based setting.

The special Euclidean group $SE(2)$. $SE(2) = SO(2) \ltimes \mathbb{R}^2$ is the group of rigid motions of the plane: an element (R_α, t) rotates spatial coordinates by angle α and translates by $t \in \mathbb{R}^2$. Unlike D_4, which is restricted to multiples of 90°, $SE(2)$ covers arbitrary rotation angles, making it more general but also harder to implement exactly in discrete image processing.

5 Method

5.1 Siamese Framework

The detector follows the siamese architecture introduced by Bromley et al. [10] for signature verification. Two branches share an identical encoder φ_θ, processing the query image x_1 and the candidate image x_2 in parallel to produce embeddings $z_1 = \varphi_\theta(x_1)$ and $z_2 = \varphi_\theta(x_2)$. A fusion module h_ψ then combines the two embeddings into a scalar score, which is passed through a sigmoid to yield the final plagiarism probability. The full model (already introduced in §3) is

$$f_\theta(x_1, x_2) = \sigma\big(h_\psi(\varphi_\theta(x_1),\ \varphi_\theta(x_2))\big). \tag{5}$$

Weight sharing between the two branches guarantees that x_1 and x_2 are mapped to the same feature space, which is a prerequisite for any meaningful pairwise comparison: without shared weights the two embeddings live in unrelated spaces and the fusion module would be comparing incommensurable vectors.

5.2 Encoders

ViT-L/16 (augmented). The baseline encoder is a standard ViT-L/16 [1]. Each 224×224 input is divided into $14 \times 14 = 196$ non-overlapping 16×16 patches; each patch is linearly projected to $\mathbb{R}^{1024}$ and the resulting sequence, together with a prepended `[CLS]` token, is processed by 24 transformer blocks with multi-head self-attention. The `[CLS]` token after the final layer norm serves as the image embedding $z \in \mathbb{R}^{1024}$. This encoder carries no equivariance guarantee. During training it receives both photometric augmentations (block P) and geometric augmentations (block G), as described in §5.5.

ViT-L/16 (baseline). Architecturally identical to ViT-L/16 (augmented), but trained with block P augmentations only; block G geometric transforms are withheld. This configuration serves as the key ablation: if this encoder matches the performance of equivariant encoders, then geometric augmentation alone suffices and architectural inductive bias adds nothing beyond what augmentation already provides.

Octic ViT (D_4-equivariant). Nordström et al. [5] replace each standard ViT layer with a D_4-equivariant variant. By Schur's lemma (§4), equivariant linear maps act independently on each irreducible component; consequently, the embedding space decomposes as

$$\mathbb{R}^d \;\cong\; \underbrace{(\mathbb{R}^{d/8})_{A_1} \oplus (\mathbb{R}^{d/8})_{A_2} \oplus (\mathbb{R}^{d/8})_{B_1} \oplus (\mathbb{R}^{d/8})_{B_2}}_{\text{four 1-dim irreps}} \oplus \underbrace{(\mathbb{R}^{d/4})_E}_{\text{one 2-dim irrep}} \,. \tag{6}$$

Invariant pooling projects onto the trivial component A_1, yielding a D_4-invariant output z. This encoder is trained with block P augmentations only.

Shift-Equivariant ViT ($\mathbb{Z}^2$-equivariant). Rojas-Gomez et al. [6] enforce exact cyclic shift-equivariance by replacing four ViT modules with adaptive polyphase variants: an A-token module (per-sample polyphase offset selection), A-WSA (adaptive window self-attention partition), A-PMerge (polyphase downsampling), and A-RPE (circular relative position encoding). These four modifications together achieve exact shift-equivariance without introducing any additional learnable parameters. Trained with block P augmentations only.

Harmformer ($SE(2)$-equivariant). Karella et al. [7] combine circular harmonic filters (steerable feature maps) with transformer self-attention blocks, achieving equivariance to continuous roto-translations. Unlike Octic ViT, which handles only 90° multiples of the dihedral group, Harmformer handles arbitrary rotation angles, providing a strictly more general symmetry guarantee at the cost of higher computational overhead. Trained with block P augmentations only.

All encoders are pre-trained on ImageNet-1K using the DINOv2 [8] self-supervised recipe.

5.3 Fusion Module

The fusion module combines two embeddings $z_1, z_2 \in \mathbb{R}^d$ into a scalar via a two-layer MLP applied to element-wise features:

$$h_\psi(z_1, z_2) = W_2\, \mathrm{ReLU}\big(W_1\, [|z_1 - z_2| \parallel z_1 \odot z_2] + b_1\big) + b_2, \tag{7}$$

where $|\cdot|$ denotes element-wise absolute value, $\odot$ element-wise product, and $[\cdot\|\cdot]$ concatenation.

Proposition 1 (Symmetry of the fusion module)**.** $h_\psi(z_1, z_2) = h_\psi(z_2, z_1)$.

Proof. $|z_1 - z_2| = |z_2 - z_1|$ and $z_1 \odot z_2 = z_2 \odot z_1$; therefore the concatenated input to the MLP is the same under transposition. □

5.4 L^2 Contrastive Regularizer

The contrastive regularizer (introduced in §3) is

$$\mathcal{L}_{\text{contr}}^{L^2}(z_1, z_2, y) = y\,\|z_1 - z_2\|_2^2 + (1-y)\,\max\bigl(0,\, m - \|z_1 - z_2\|_2\bigr)^2. \tag{8}$$

We use L^2 distance rather than cosine similarity for a principled reason. As established by Hadsell et al. [14], the L^2 contrastive formulation is motivated by its geometric properties: $\|z_1 - z_2\|_2$ is an *isometric invariant*, preserved by any orthogonal transformation of the embedding space. This makes it naturally compatible with G-invariant encoders that may differ from one another only by an isometry of $\mathbb{R}^d$. Cosine similarity, by contrast, is scale-dependent — it is not preserved by orthogonal maps that mix the direction and magnitude of vectors — and therefore does not share this property.

Proposition 2 (Isometry invariance of $\mathcal{L}_{\text{contr}}^{L^2}$)**.** *Let $U\colon \mathbb{R}^d \to \mathbb{R}^d$ be an orthogonal map. Then $\mathcal{L}_{\text{contr}}^{L^2}(Uz_1, Uz_2, y) = \mathcal{L}_{\text{contr}}^{L^2}(z_1, z_2, y)$.*

Proof. $\|Uz_1 - Uz_2\|_2 = \|U(z_1 - z_2)\|_2 = \|z_1 - z_2\|_2$, since orthogonal maps preserve the L^2 norm. □

For each encoder we compare two conditions: $\lambda > 0$ (with L^2 regularizer) and $\lambda = 0$ (without).

5.5 Augmentation Blocks

Block P (photometric augmentations) is applied to all encoders during training:

Transform	**Parameters**
ColorJitter	brightness=0.2, contrast=0.2, saturation=0.1, hue=0.05
RandomGrayscale	$p = 0.05$
GaussianBlur	kernel=3, $p = 0.1$
ImageNet normalization	$\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$

Block G (geometric augmentations) is applied only to ViT-L/16 (augmented):

Transform	**Parameters**
RandomHorizontalFlip	$p = 0.5$
RandomVerticalFlip	$p = 0.5$
RandomChoice of rotation	$\{0°, 90°, 180°, 270°\}$
RandomResizedCrop	scale=$(0.7, 1.0)$, ratio=$(0.9, 1.1)$

Equivariant encoders guarantee the same embedding for any $g \in G$ applied to the input by design, so adding block G augmentation for them would produce zero-gradient pairs for the head — not harmful, but redundant. The core experimental hypothesis is therefore that block G can be dropped for equivariant encoders without loss of detection performance.

5.6 Theoretical Properties: Invariance Propagation

Proposition 3 (Invariance propagation)**.** *If φ_θ is G-invariant, then $f_\theta(g \cdot x_1, g \cdot x_2) = f_\theta(x_1, x_2)$ for all $g \in G$ and all $x_1, x_2 \in \mathcal{X}$.*

Proof. G-invariance of φ_θ gives $\varphi_\theta(g \cdot x_i) = \varphi_\theta(x_i)$ for $i = 1, 2$. Since h_ψ and σ are deterministic functions of their inputs, the output of f_θ is unchanged. □

This guarantees that the model's plagiarism score is the same regardless of which $g \in G$ is applied to the input pair, matching the invariance required for geometric plagiarism detection.

6 Experiments

6.1 Dataset and Pre-training

All encoders are pre-trained on ImageNet-1K using the DINOv2 [8] self-supervised recipe. Pre-training is performed separately for each encoder architecture where DINOv2-compatible weights exist; for architectures where pre-trained weights are not publicly available, we initialize from a standard ViT-L/16 DINOv2 checkpoint and adapt.

We use DomainNet [15] as the evaluation benchmark. DomainNet contains images from six visually diverse domains: real photograph, painting, clipart, quickdraw, infograph, and sketch. We construct a balanced test set of 1200 pairs: 100 positive pairs and 100 negative pairs for each of the six domains (600 positives and 600 negatives in total). Each image is resized to 224×224 pixels, yielding a feature count of $3 \times 224 \times 224 = 150{,}528$ scalar features per image. Positive pairs consist of a source image and a transformed copy; negative pairs consist of two distinct images from the same domain (hard negatives).

6.2 Training Pipeline

The experiment comprises 10 training runs: 5 encoders $\times$ $\{\lambda = 0$ (without L^2 regulariser), $\lambda > 0$ (with L^2 regulariser)$\}$.

For each run, we load base images from a training split and construct pairs via the cross-batch strategy of Dorin et al. [9]: a batch of B images is augmented to form two views per image (block P for all encoders, additionally block G for ViT-L/16 augmented), and all B^2 pairwise combinations are formed with labels $y_{ij} = \mathbb{1}[i = j]$. Embeddings are computed through the frozen encoder and fusion module, the composite loss $\mathcal{L}$ is evaluated, and AdamW updates the fusion parameters ψ. All hyperparameters are listed in Table 2.

Table 2: Training hyperparameters.

Parameter	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-2}
LR scheduler	MultiStepLR, milestones $[5, 10]$, $\gamma = 0.5$
BCE weights	$w_+ = 0.3$, $w_- = 0.7$
Epochs	15
Contrastive margin m	1.0
Contrastive weight λ (when enabled)	0.1
Batch size B	32

6.3 Evaluation Protocol

The evaluation workflow for each (encoder, $\pm L^2$) run is: train on the training split, select a decision threshold on the validation fold to maximise F1, then compute final metrics on the held-out test set.

The primary metrics are false positive rate (FPR) and Recall, both stratified by transformation class: ColorJitter (CJ), Gaussian Noise (GN), Grayscale (GS), R90, R180, R270, Watermark (WM), and their pairwise combinations. FPR is reported for direct comparability with Dorin et al. [9]. Secondary metrics include the F1 score and a t-SNE visualisation of encoder embeddings computed on the full DomainNet evaluation set, with the six domains rendered as distinct point colours.

6.4 Code Organisation

The implementation targets the following modular structure, with all hyperparameters centralised in a single configuration file:

- `code/main.py` — single entry point for training and evaluation

- `code/encoders/` — encoder wrappers (ViT-L/16, Octic ViT, Shift-Equivariant ViT, Harmformer)
- `code/siamese/` — siamese network and fusion module
- `code/data/` — dataset loading and augmentation blocks P and G
- `code/train/` — training loop and loss computation
- `code/eval/` — evaluation, threshold selection, metric computation, t-SNE
- `code/config.yaml` — centralised hyperparameters (all values from Table 2)

The current baseline is implemented in `code/image-plagiarism-detection-with-eq-encoders.ipynb`, which serves as the reference for hyperparameters and the cross-batch pairing strategy.

6.5 Planned Figures and Tables

All figures are to be produced in `.pdf` or `.eps` format at 300 DPI, with fonts at 12–16 pt and lines at 2 pt width.

- **Figure 1.** Siamese architecture schema: encoder branches, fusion module, sigmoid head. (Axes: not applicable.)
- **Figure 2.** Diagrams of D_4, $\mathbb{Z}^2$, $SE(2)$ with example actions on a sample image. (Axes: spatial coordinates of the image under each transformation.)
- **Figure 3** (source data plot). Distribution of DomainNet evaluation pairs. x-axis: domain name; y-axis: pair count.
- **Figure 4** (main message). FPR by encoder and regularisation condition. x-axis: encoder name; y-axis: FPR; grouped bars: $\lambda = 0$ vs. $\lambda > 0$.
- **Figure 5** (main message). Recall stratified by transformation class. x-axis: transformation class (CJ, GN, GS, R90, R180, R270, WM, combos); y-axis: Recall; bars grouped by encoder.
- **Figure 6.** t-SNE of encoder embeddings on DomainNet. Five subplots (one per encoder); point colours: six DomainNet domains.

In addition to Table 1 (§ 1) and Table 2 (§ 6):

- **Table 3.** Main results: FPR for each (encoder, regularisation) run.
- **Table 4.** Recall stratified by transformation class for each (encoder, regularisation) run.

6.6 Results

7 Conclusion

We presented a siamese plagiarism detector that benchmarks five encoder configurations spanning three symmetry groups—D_4 (Octic ViT), $\mathbb{Z}^2$ (Shift-Equivariant ViT), and $SE(2)$ (Harmformer)—against a plain ViT-L/16 baseline. By fixing the DINOv2 pre-training protocol and varying only the geometric inductive bias and the presence of an L^2 contrastive regularizer, the framework isolates the contribution of built-in G-invariance from that of data augmentation across 10 runs. The theoretical foundation (§ 5) establishes three guarantees: symmetry of the fusion module, isometry invariance of the L^2 contrastive loss, and full invariance propagation through the siamese pipeline. The central hypothesis—that G-invariant encoders trained without block G augmentations match or exceed a ViT-L/16 trained with full augmentation—remains subject to the experimental evaluation on DomainNet.

The experimental runs (§ 6.6) will be completed in subproject C: implementing Shift-Equivariant ViT and Harmformer, pre-training all five encoders on ImageNet-1K, and executing the 10 siamese fine-tuning experiments. Results will be reported as FPR and Recall stratified by transformation class on the balanced DomainNet test set. We anticipate that the equivariant encoders will validate the inductive bias hypothesis, reducing reliance on geometric augmentation while preserving detection performance.

References

[1] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[2] Boyuan Chen, Zhuo Xu, Sean Kirmani, Brain Ichter, Dorsa Sadigh, Leonidas Guibas, and Fei Xia. SpatialVLM: Endowing vision-language models with spatial reasoning capabilities. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14455–14465, 2024.

[3] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.

[4] Taco S. Cohen and Max Welling. Group equivariant convolutional networks. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, volume 48 of *JMLR: W&CP*, pages 2990–2999, 2016.

[5] David Nordström, Johan Edstedt, Fredrik Kahl, and Georg Bökman. Octic vision transformers: Quicker vits through equivariance, 2025. URL `https://arxiv.org/abs/2505.15441`.

[6] Renan A. Rojas-Gomez, Teck-Yian Lim, Minh N. Do, and Raymond A. Yeh. Making vision transformers truly shift-equivariant, 2023. URL `https://arxiv.org/abs/2305.16316`.

[7] Tomáš Karella, Adam Harmanec, Jan Kotera, Jan Blažek, and Filip Šroubek. Harmformer: Harmonic networks meet transformers for continuous roto-translation equivariance, 2024. URL `https://arxiv.org/abs/2411.03794`.

[8] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

[9] Daniil Dorin, Kseniia Varlamova, and Andrey Grabovoy. Pairwise image matching for plagiarism detection, 2024. URL `https://github.com/DorinDaniil/Pairwise-Image-Matching`. Preprint; training protocol and augmentation details.

[10] Jane Bromley, Isabelle Guyon, Yann LeCun, Eduard Säckinger, and Roopak Shah. Signature verification using a "Siamese" time delay neural network. In *Advances in Neural Information Processing Systems (NIPS)*, volume 6, 1993.

[11] Yan Ke, Rahul Sukthankar, and Larry Huston. Efficient near-duplicate detection and sub-image retrieval. In *Proceedings of the ACM International Conference on Multimedia*, pages 869–876, 2004.

[12] Maurice Weiler and Gabriele Cesa. General E(2)-equivariant steerable CNNs. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

[13] Sumit Chopra, Raia Hadsell, and Yann LeCun. Learning a similarity metric discriminatively, with application to face verification. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 539–546, 2005.

[14] Raia Hadsell, Sumit Chopra, and Yann LeCun. Dimensionality reduction by learning an invariant mapping. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1735–1742, 2006.

[15] Xingchao Peng, Qinxun Bai, Xide Xia, Zijun Huang, Kate Saenko, and Bo Wang. Moment matching for multi-source domain adaptation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1406–1415, 2019.